

Ahad Kamal

Software Engineer

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Software Engineer with strong skills and relevant experience in c++ and Unreal, debugging, and working in large teams across disciplines. Highly motivated, flexible, and professional with solid communication skills

Team Project Experience

Animation and Systems Programmer - *Hamsterballin* (Unreal)

January - May 2026

- Collaborated with artists to develop the logic behind the game's animation system and created a system for swapping character costumes and colors that other programmers and artists could use to streamline the process of implementing them.
- Implemented the game's camera system which allowed for camera switching and setting updates all during runtime, as well as functionality for various obstacles that would either benefit or harm the player in interesting and fun ways.
- Debugged several issues that lead to lower performance by using Unreal Insights performance profiling tools and improved the game's overall performance by 25%

Gameplay Programmer - *Gravebound* (Unreal)

August 2024 – May 2025

- Developed several core mechanics: movement, weapons, upgrades, and enemies alongside a team of five additional programmers, resulting in a unique 3rd person Roguelite game.
- Collaborated concurrently with a team of over 30 programmers, designers, artists, and audio producers through the use of Perforce's file check out system.

Lead Technical Director - *The Kraken's Tooth* (Maya)

August 2024 – May 2025

- Responsible for managing the animation pipeline and ensuring that production runs smoothly across all departments consisting of around 40 artists.
- Created tools that aided artists and investigated issues that would hinder development throughout the early stages of the production.

Personal Projects

Doomenstein (Custom C++ Engine)

January - May 2026

- Programmed the entire framework for a Doom/Wolfenstein clone, including the following gameplay systems: physics based movement, split-screen multiplayer, player and enemy AI controllers, and data driven xml file parsing.
- Converted my existing OpenGL rendering pipeline with a DirectX11 pipeline, with support for both 2D and 3D rendering, textures, shaders, sprite sheets, and sprite animations.

NetChess (Custom C++ Engine)

May - July 2026

- Recreated the classic game of chess in my engine that accounts for piece movement rules, capturing logic, and proper turn order.
- Coded a manual obj file parser that reads in vertex, UV, and normal data and then outputs the data into a complete 3D model, with additional supporting functions that can create normals, tangents, and bitangents for models that do not contain this information.
- Implemented a more sophisticated BlinnPhong shader that supports multiple directional lights, point lights, spotlights, specular, glossiness, emissivity, and several debug shading modes that isolate each of these lights and values
- Implemented the third party ImGui library for easy UI creation and implementation, including UI functionality for controlling light values and debug modes

FBXModelViewer (Custom C++ Engine)

May - July 2026

- Created a system for loading and rendering fbx files using the Autodesk FBX SDK.
- Coded a pipeline for importing fbx files involving retrieving the model's vertices, uvs, normals, joints, and animation data, then structuring the data into custom classes that serve as wrappers for easier use.
- The end result being a fully rendered, textured, lit model, with animation keyframes being calculated using the bone hierarchy on the CPU and the mesh then skinned and animated on the GPU.
- Implemented a custom .model file type that gets baked the first time the model is loaded, which is then instead parsed and loaded in manually without having to re-retrieve the data from the fbx file. This resulted in a 60% speed increase on load times

Core Skills

Programming Languages: C#, C++, Python, Blueprints, Java, and Haskell

Software: Unity, Maya, Unreal Engine, Perforce, Windows, WSC, WVS, Blender, Github, and Microsoft Office

Soft Skills: Agile Development, Object Oriented Programming, Functional Programming, Mathematics, Flexibility, Communication, Organization, Receiving Feedback, Problem-Solving, Responsibility, and Integrity

Work Experience

Unity Instructor - UTD CS Outreach

June – August 2024

- Taught students K-12 a Unity summer course involving learning about various features and systems behind the Unity engine and guided them through the process of using the editor and coding in C# with the end goal of creating a game in Unity.

Education

Masters of Interactive Technology
Bachelor of Science in Computer Science

Southern Methodist University Guildhall **2027**
The University of Texas at Dallas **2025**

Relevant Coursework: Professional Communication, Data Structures, Database Systems, Programming Languages, Automata Theory, Digital Logic, Operating Systems, Computer Animation, Computer Graphics, Game Lab, and Animation Lab